Bence Soóki-Tóth

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Research Interest

My research interests lie at the intersection of cryptography and game theory, focusing on privacy-enhancing technologies and the security of decentralized systems. I am broadly interested in developing efficient protocols with formal security guarantees that withstand rational behavior and collusion facilitated through smart contracts.

Education

Aarhus University — Aarhus, Denmark
Master of Computer Science, Specialization in Cryptography — *Aug 2025 – Jun 2027*
- **Coursework:** Cryptology, Cryptologic Protocol Theory, Succinct Proof Systems

Eötvös Loránd University — Budapest, Hungary
Bachelor of Computer Science, Mathematical Modelling Specialization — *Sep 2022 – June 2025*
- **Thesis:** *Efficient Calculation of Permanent Function Gradients in Photonic Quantum Computing Simulations*
- **Coursework:** Discrete Mathematics I, II; Analysis I, II, III; Probability Theory; Statistics

Aalto University — Espoo, Finland
Exchange Semester — *Aug 2024 – Dec 2024*

Publications

Bribers, Bribers on The Chain, Is Resisting All in Vain? Trustless Consensus Manipulation Through Bribing Contracts [pdf] — *FC'26*
Bence Soóki-Tóth*, István András Seres*, Kamilla Kara, Ábel Nagy, Balázs Pejó, Gergely Biczók (*equal contribution)*
Demonstrates how mutually distrusting parties can coordinate bribery on Ethereum's consensus protocol.

Preprints

Private Delegation of (Non-)Membership Proof Updates in Cryptographic Accumulators [pdf]
Bence Soóki-Tóth*, Botond Glasz, Alireza Kavousi, István András Seres*
Novel protocols for privately delegating accumulator proof updates, with game-based security proofs and $O(1)$ online client computation *i.e.,* independent of the number of updates.

Research Experience

Research Assistant — May 2025 – Jul 2025
Eötvös Loránd University — *Budapest, Hungary*
- Designed and developed the encryptor module of QCIHungary, the Hungarian national contribution to the EU's EuroQCI quantum communication infrastructure.
- Implemented encryption endpoints and caching for fault-tolerant recomputation, reducing restart overhead.

Student Research Fellow — Feb 2025 – Apr 2025
ELTE-OTP Cybersecurity Industrial Laboratory — *Budapest, Hungary*
- One of 8 students selected for a research fellowship to work on problems proposed by OTP Bank.
- Analyzed the Hungarian DLT-based real estate infrastructure.

Invited and Contributed Talks

Trustless Consensus Manipulation Through Bribing Contracts
- BSides Aarhus 2026, Denmark [slides] — 20 Jun 2026
- Aarhus Crypto Seminar, Denmark [slides] — 19 Mar 2026
- Financial Cryptography and Data Security 2026, St. Kitts and Nevis [slides] — 02 Mar 2026

Private Delegation of (Non-)Membership Proof Updates in Cryptographic Accumulators
- Nordicrypt 2026, Sweden — 30 Oct 2026

Teaching and Academic Service

Teaching Assistant, Software Security and Research Methodology, Aarhus University — Aug 2026 – Dec 2026

Conference volunteer, TPMPC 2026, Aarhus University — May 2026

Technical Mentor, Bachelor's Thesis Project
Eötvös Loránd University (Official supervisor: István András Seres) — Nov 2025 – Apr 2026

- Advised a bachelor's student on the implementation of the RSA accumulator in Rust.

Grants, Awards and Scholarships

Academic Merit Scholarship, Eötvös Loránd University — multiple semesters — 2022 – 2025

unitaryHACK 2025 — 11th globally, Unitary Foundation — May 2025

HSUP National Finalist and Scholarship Recipient, Hiraeth Labs, one of 15 finalist teams selected from 178, Hungarian Innovation Agency — 2024

Projects and Open-Source Contributions

private-accumulator-proof-delegation (source) | Rust — Nov 2025 – Apr 2026

- Implemented NIZKs, the bilinear and class-group accumulator. Artifact for the PKC'27 submission.
- Optimised bilinear accumulator operations via synthetic division, reducing overhead by $\approx 30\%$.

reth (source) | Rust — Nov 2025 – Jan 2026

- Implemented batching for multiproof messages, and associated tests.

bribery-zoo (source) | Rust, Solidity — Mar 2025 – Sep 2025

- Designed and implemented trustless bribing contracts against Ethereum's consensus protocol. Artifact for the FC'26 paper.

permanent-boost (source) | C++, CUDA — Jun 2024 – Jun 2025

- Derived a multiplicity-aware gradient algorithm whose complexity depends exponentially on mode occupancy rather than photon count.
- Outperformed the state-of-the-art Piquasso backend from $n \geq 11$ upward, now integrated into Piquasso.

Industry Experience

Co-founder — Feb 2024 – Jun 2024
Hiraeth Labs — *Budapest, Hungary*

- Co-founded a student venture to develop AI-assisted explanations of laboratory results, collaborating with software developers and clinicians.

Test Automation Trainee — Jul 2023 – May 2024
SAP — *Budapest, Hungary*

- Extended the Appium-based desktop test automation framework for SAP Service and Asset Manager, raised end-to-end coverage of field service workflows from 5% to 88%.

Full-stack Developer — Jun 2022 – Oct 2022
Coding Sans — *Budapest, Hungary*

- Built Node.js/TypeScript backend services for a SaaS form-builder, designed caching and load-balancing for analytics endpoints.

Technical Skills and Languages

Programming: Rust, C++, CUDA, Solidity, Coq (Rocq)

Cryptography: zk-SNARKs, cryptographic accumulators, game-based security

Technologies: Trusted Execution Environments (TEEs), formal verification

Languages: Hungarian (native), English (C1), Danish (beginner)